Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
Department of Computer Science and Engineering
Academic Year 2026-27

Mini Project - Weekly Progress Report

Mini Project Title: Persistent Event-Driven Memory Cache Module

Name of Students:

1. Jay Umesh Shinde (2024800115)
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3. Nikhil Shelar (2024800113)

Name of the Supervisor: Prof. Suhas Kakade

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
04/08/2026 - 10/08/2026	**Week 2 - Related Systems Study and Detailed Design** • Studied related production ideas such as Redis-style in-memory caching/eviction and database-style WAL, transactions and crash recovery. • Refined the implementation stack to Java 17 with no mandatory external Java dependencies for easy college-lab execution. • Designed versioned WAL records with monotonic sequence numbers, CRC32 integrity checks and configurable durability. • Planned ordered range queries using a real B-tree, TTL expiry, compare-and-set version checks, per-key transaction locking and snapshot recovery. • Designed event-driven observability using an ordered event bus, metrics projection, JSONL audit logging and a recent-event journal.	1. _____ 2. _____ 3. _____	To be filled by Supervisor	
Next Week Plan	**Planned Work** • Integrate the core implementation and recovery flow. • Prepare verification evidence and Phase I presentation.		To be filled by Supervisor	